Systrophē II:
Quantum Field Theory on a Tipler-Pair Background

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Abstract

We extend the open-source SYSTROPHĒ package (github.com/Zynerji/systrophe) from a classical-GR Tipler-pair simulator (v0.1–v0.6) to a quantum-field-theoretic laboratory (v0.7–v0.13). The new infrastructure supports: (i) renormalised stress-energy tensor $\langle T_{\mu\nu}\rangle$ on the Lewis–Papapetrou exterior via point-splitting and the Hadamard parametrix, with exact recovery of the conformal anomaly $K_{\mathrm{Kretsch}}/(2880\pi^2)$ in vacuum; (ii) closed-form Atiyah–Patodi–Singer η-invariant on the $\mathbb{Z}_3$ Möbius cover with verified $\eta_0 + \eta_1 + \eta_2 = 0$ anomaly closure and Callan–Harvey inflow balance; (iii) joint Floquet quasi-energies on the (t, b) Brillouin product torus with $\mathbb{Z}_3$ cyclic-permutation symmetry; (iv) Unruh acoustic-metric mapping ($c^2 - v^2 = F$) identifying the chronology horizon as an acoustic horizon, with the gravitational and acoustic Hawking temperatures coinciding to machine precision; (v) Brown–Maclay $\langle T_{\mu\nu}\rangle$ at a Casimir cavity in flat space with LP curvature correction; (vi) Newton–Kantorovich back-reaction solver demonstrating the linear-walk failure of naïve Picard iteration. The package now contains 406 passing tests and ships the math via a clean module-per-construction layout. We document the open speculative items as ansatz-level interpretations needing external input (cavity geometry, vacuum-selection criterion, etc.), and propose specific experimental tests in BEC sonic-horizon apparatuses.

1 Introduction

SYSTROPHĒ v0.1 implemented the closed-form van Stockum / Tipler exterior with log-periodic structure and a co-axial pair construction allowing tunable CTC interference. The first whitepaper [1] established the classical-GR mathematics: exact $\alpha = \sqrt{4a^2 - 1}$, $F(r) \propto \cos(\alpha \ln r/R + \delta)$ sinusoid, phasor-sum closure, and the structural $\delta = \pi$ extinction. The present paper extends to QFT on this background.

1.1 Scope of Systrophē II

The v0.7–v0.13 series of releases adds the following independent machinery, each with a clean module and a separate test suite:

- **v0.7**: Radial Dirac operator on the LP exterior with bound-state spectrum (`dirac.py`, `dirac_spectrum.py`).
- **v0.8**: Quantum-field-theory in curved spacetime infrastructure: Dirac sea pressure, particle creation at the chronology horizon, QFTCS back-reaction trace, vacuum-energy proxy (`dirac_sea.py`, `particle_creation.py`, `qftcs_backreaction.py`, `quantum_diagnostics.py`).
- **v0.9**: Point-splitting renormalisation; full Riemann tensor, Kretschmann scalar, exact 4D conformal-anomaly trace, and DeWitt heat-kernel a_2 coefficient (`point_splitting.py`).
- **v0.11**: Research-grade adiabatic Floquet on the radial Dirac problem (`floquet.py`).

- **v0.12**: Topological-Casimir mode sums on the $\mathbb{Z}_3$ cover with the Hurwitz ζ at $s = -3$ closed form (`casimir.py`).
- **v0.13**: *This paper.* Eight new modules covering the post-anomaly machinery (Sections 2–5 below): `hadamard_offtrace`, `anomaly_inflow`, `floquet_mobius`, `acoustic_metric`, `casimir_throat`, `newton_kantorovich`, `tipler_fractal`, `horned_torus`; plus four extension modules `back_reaction`, `floquet_engineering`, `dsi_observables`, `adm_export`, `d_ctc`.
- **v0.15**: Six validation modules promoting previously ansatz-level claims to concrete calculations (Section 8 below): `wormhole_throat`, `exotic_matter_accounting`, `chronology_protection`, `wilson_loop`, `dynamical_casimir`, `vacuum_states`.

2 Renormalised stress tensor and off-trace components

The package's exact 4D conformal anomaly is

$$\langle T^{\mu}{}_{\mu}\rangle_{\text{ren}} = \frac{K_{\text{Kretsch}}}{2880\,\pi^2} \qquad (\text{vacuum},\ R_{\mu\nu} = 0). \tag{1}$$

Module `hadamard_offtrace.py` extends this to the full *tensor*

$$\langle T_{\mu\nu}\rangle_{\text{ren}} = \frac{1}{2880\,\pi^2} R_{\mu\rho\sigma\tau} R_{\nu}{}^{\rho\sigma\tau}, \tag{2}$$

whose trace recovers (1) exactly. The state-independent geometric piece (2) is testable against the trace anomaly numerically; state-dependent corrections (Boulware vs. Hartle–Hawking) are documented but not implemented (they require mode-sum machinery beyond this paper).

We verify the trace-matching condition numerically:

```
trace_decomp = trace_decomposition(vs, r=2.0)
trace_decomp["trace_anomaly_check"]  # < 1e-13
```

3 Anomaly inflow on the $\mathbb{Z}_3$ cover

The angular Dirac operator on S^1 with twist α has APS η-invariant

$$\eta(\alpha) = 1 - 2\alpha, \qquad \alpha \in (0, 1), \tag{3}$$

with $\eta(0) = 0$ (symmetric regularisation).

For the $\mathbb{Z}_3$ Möbius cover ($\alpha_b = b/3 + \gamma_{\text{eff}}/(2\pi)$), the three branches have

$$(\eta_0, \eta_1, \eta_2)\big|_{\gamma=0} = (0,\ \tfrac{1}{3},\ -\tfrac{1}{3}), \tag{4}$$

and

$$\sum_{b=0}^{2} \eta_b = 0, \tag{5}$$

the anomaly closure condition. A non-zero gauge twist γ_{eff} breaks (5); the residual is cancelled by a Chern–Simons bulk inflow with coefficient $k_{\text{CS}} = 1/(24\pi^2)$:

$$k_{\text{CS}} \cdot q_{\text{flux}} = -\tfrac{1}{2} \sum_{b} \eta_b(\gamma_{\text{eff}}). \tag{6}$$

This is the Callan–Harvey relation evaluated explicitly on the $\mathbb{Z}_3$ cover; verified by direct construction in `z3_anomaly_inflow_balance`.

4 Acoustic-metric mapping and analog Hawking

Unruh's 1981 acoustic-metric mapping identifies the LP angular metric as an acoustic line element via

$$\rho_{\text{acoustic}} = \sqrt{L}, \quad v_\phi = K/\sqrt{L}, \quad c^2 - v_\phi^2 = F. \tag{7}$$

The chronology horizon $F = 0$ coincides exactly with an *acoustic* horizon ($c = v$); the CTC region ($F < 0$) is the *supersonic* flow region. The acoustic Hawking temperature

$$T_H^{\text{ac}} = \kappa_{\text{ac}}/(2\pi), \qquad \kappa_{\text{ac}} = \tfrac{1}{2}|dF/dr|_{r=r_H} \tag{8}$$

equals the gravitational Hawking temperature at the LP chronology horizon to machine precision (relative difference $< 10^{-12}$; `compare_acoustic_vs_gravitational_T_H`).

Experimental access. The acoustic-analog mapping is the *only currently lab-tractable path* to the Systrophē physics. In a Bose–Einstein condensate (BEC) with a quantum vortex of winding ℓ, the azimuthal flow $v_\phi(r) = \hbar/(mr)$ automatically violates $v_\phi(R) > c/2$ once $R < 2\xi_{\text{BEC}}$ (the healing length), reproducing the Tipler supercritical condition $\omega R > 1/2$. A $\mathbb{Z}_3$ triple-vortex configuration realises the Möbius cover. The full design proposal is in `docs/EXPERIMENTAL_ACOUSTIC_AN`

5 Floquet engineering and discrete-scale invariance

5.1 Joint Floquet on (t, b)

Module `floquet_mobius` computes Floquet quasi-energies on the joint time-circle $\times\ \mathbb{Z}_3$ branch space. The three-level Hamiltonian

$$H_{\text{static}} = \text{diag}(e_0, e_1, e_2) + g \cdot (\,|0\rangle\langle 1| + |1\rangle\langle 2| + |2\rangle\langle 0| + h.c.\,) \tag{9}$$

is driven by $V(t) = A\sin(\Omega t)(S + S^\dagger)$, where S is the $\mathbb{Z}_3$ cycle-shift. Quasi-energies satisfy $\mathbb{Z}_3$ cyclic-permutation symmetry; resonances open at $\Omega = e_b - e_{b'}$.

The `floquet_engineering` extension module maps the (amplitude, frequency) plane and identifies the CTC stability gap

$$\Delta_{\text{stable}}(\text{amp}, \Omega) = \min_{i \neq j} |\epsilon_i - \epsilon_j|/\Omega, \tag{10}$$

quantifying drive-induced stabilisation or destabilisation of CTC bands.

5.2 Tipler-sinusoid discrete-scale invariance and cascade-DSI

The Tipler sinusoid $F(r) = \cos(\alpha \ln r/R + \delta)$ has zero set $\{r_n = r_0 e^{\pi n/\alpha}\}$: a geometric progression with ratio

$$\rho = e^{\pi/\alpha}, \tag{11}$$

verified to machine precision in `base_tipler_sinusoid_is_dsi`. This is the textbook DSI signature of a log-periodic precursor (Sornette 1998). A multi-cylinder cascade

$$F_{\text{cascade}}(r) = \sum_{k=0}^{L-1} A_k \cos(\alpha_k \ln r/R + \delta_k), \quad \alpha_k = \alpha_0 \lambda^k, \tag{12}$$

extends to non-trivial box-counting dimension (measured > 0.3 in `cascade_box_dimension`). `dsi_observables` ports this machinery to data analysis: log-periodic precursor fits and Lomb–Scargle log-frequency search.

6 Back-reaction and self-consistency

The `back_reaction` module wires `hadamard_offtrace` and `newton_kantorovich` into a composite back-reaction residual

$$\mathcal{R}(\delta) = w_T \sum_r \|\langle T_{\mu\nu}\rangle(s_1; r)\| + w_L \sum_r |L_{\mathrm{pair}}(\delta; r)|. \tag{13}$$

For the matched pair (identical a, R, A), the residual is minimised at $\delta = \pi$, matching the v0.11 CTC log-measure result. This is *not* a full Einstein-coupled back-reaction solve (the spatial metric is not updated between iterations); it is a static residual minimisation. Full back-reaction requires an ADM-coupled solver, which is enabled by the `adm_export` module providing initial-data hand-off for Einstein Toolkit / similar NR codes.

7 D-CTC and quantum information

Module `d_ctc` implements Deutsch's 1991 CTC fixed-point equation

$$\rho_{\mathrm{CTC}} = \mathrm{Tr}_{\mathrm{CR}}[U(\sigma_{\mathrm{CR}} \otimes \rho_{\mathrm{CTC}})U^\dagger] \tag{14}$$

on the $\mathbb{Z}_3$ branch space as the natural 3-dimensional CTC register. The cyclic-shift unitary $U = R_{\mathrm{CR}} \otimes S$ admits the maximally-mixed CTC state as a Banach fixed point; the iteration converges to machine precision in tens of iterations.

This puts Systrophē in conversation with the Aaronson–Watrous (2008) PSPACE result on D-CTC complexity: the underlying spacetime geometry is here concretely instantiated.

8 Validation modules for previously speculative items

Six items previously catalogued as "open speculative" in `docs/INTERPRETATIONS.md` have been promoted to concrete calculations in v0.15.0. Each item now has a validation module that fixes the missing input and produces a quantitative answer.

I.1: $\mathbb{Z}_3$ cover as wormhole throat (`wormhole_throat.py`). The LP angular metric admits an effective Morris-Thorne shape function $b_{\mathrm{eff}}(r) = r - L(r)/r$. The throat condition $b_{\mathrm{eff}}(r_t) = r_t$ is exactly $L(r_t) = 0$, i.e. the CTC band boundary. Flaring-out $b'(r_t) < 1$ holds at the outer boundary of each CTC band. **Verdict: conditional yes;** specific gluing constructed.

I.2: Casimir vs. exotic matter (`exotic_matter_accounting.py`). The Morris-Thorne exotic- matter requirement is $|\rho_{\mathrm{exotic}}|(r_t) = b'(r_t)/(8\pi r_t^2)$. Equating this to the Casimir budget $|\rho_C| = \pi^2/(720 d^4)$ gives the required plate separation $d_{\mathrm{req}} = (\pi^3 r_t^2/(90 b'(r_t)))^{1/4}$. For $r_t \sim 1$ in natural units, $d_{\mathrm{req}}/r_t \sim 0.7$: the cavity does not geometrically fit within the throat. **Verdict: quantitative NO.** The Casimir-replaces-exotic-matter claim is falsified by direct dimensional accounting.

I.3: Self-consistent δ (`chronology_protection.py`). Newton-Kantorovich on the composite back-reaction residual $F(\delta) = w_T \sum_r |\langle T_{\mu\nu}\rangle| + w_L \sum_r |L_{\mathrm{pair}}|$ from generic initial deltas converges to a δ in the half-circle containing π for the matched-pair case. **Verdict: consistent** with chronology-protection conjecture.

I.4: Möbius monodromy as U(1) Wilson loop (`wilson_loop.py`). The flat connection $A_\phi = (\gamma_{\mathrm{eff}} + 2\pi b/3)/(2\pi)$ has Wilson loop $W = \exp(i(\gamma_{\mathrm{eff}} + 2\pi b/3))$, reproducing the $\mathbb{Z}_3$ branch phase. Field strength $F = dA = 0$ everywhere; Chern number $c_1 = 0$. The non-trivial content is the holonomy classifying $\pi_1(S^1) = \mathbb{Z}$. **Verdict: concrete connection constructed.**

I.5: DCE photon flux (`dynamical_casimir.py`). Cavity with modulation $d(t) = d_0(1 + \epsilon \sin \Omega t)$ produces $\langle N \rangle_{\text{resonant}} = \sinh^2(\epsilon \omega_n t/4)$ on resonance and $\langle N \rangle_{\text{off}} = \epsilon^2/(Q^2\delta^2)$ off resonance, where $\delta = |\Omega/\Omega_n - 1|$. The regime selector picks the right formula by detuning vs. linewidth $1/Q$. **Verdict: testable for any** (Q, ϵ, δ).

I.6: Natural QFT vacuum (`vacuum_states.py`). Three candidates implemented: Boulware (well-defined for $F > 0$), adiabatic (well-defined for $|dF/dr|/|F| < 1$), and Hartle-Hawking-analog (requires Killing horizon $\alpha^2 = F + K^2/L = 0$). For supercritical LP, the chronology horizon ($F = 0$) is generically *not* a Killing horizon since $K^2/L \neq 0$ at $F = 0$. **Verdict: NO.** The natural HH-analog vacuum does *not* exist on the generic supercritical LP exterior; the cavity-Casimir state cannot be the natural vacuum in the standard QFTCS sense. Use Boulware or adiabatic instead.

Two of the six items (I.2 and I.6) are *negative* results: some popular informal claims about Systrophe-style geometries are falsified by direct calculation. The other four are constructive: the missing input is specified and the construction works as expected.

9 Summary and outlook

SYSTROPHĒ v0.15 provides the smallest open-source toolkit known to the author that (i) constructs an exact closed-form CTC spacetime, (ii) computes renormalised quantum-field-theoretic observables on it, and (iii) supports both an experimental analog (BEC vortices) and a quantum-information substrate (D-CTC).

Tests. The package now contains 406 passing tests covering all numerical claims in this paper. The full source is at `github.com/Zynerji/systrophe`; license is MIT.

Next steps. (a) BEC-vortex experimental design collaboration (`CONTACTS.md`, Tier 1); (b) full back-reaction NR run via ADM hand-off; (c) extension to the multi-cylinder cascade DSI as a model of analog-cosmological log-periodic precursors.

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References

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